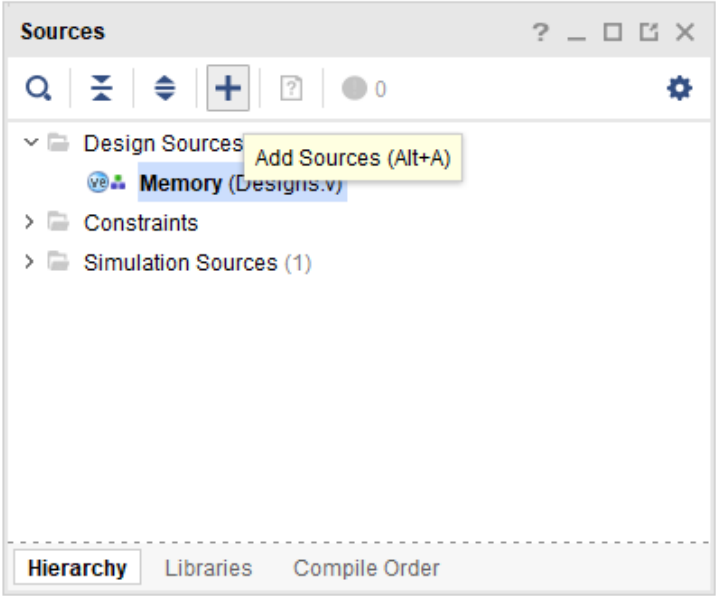
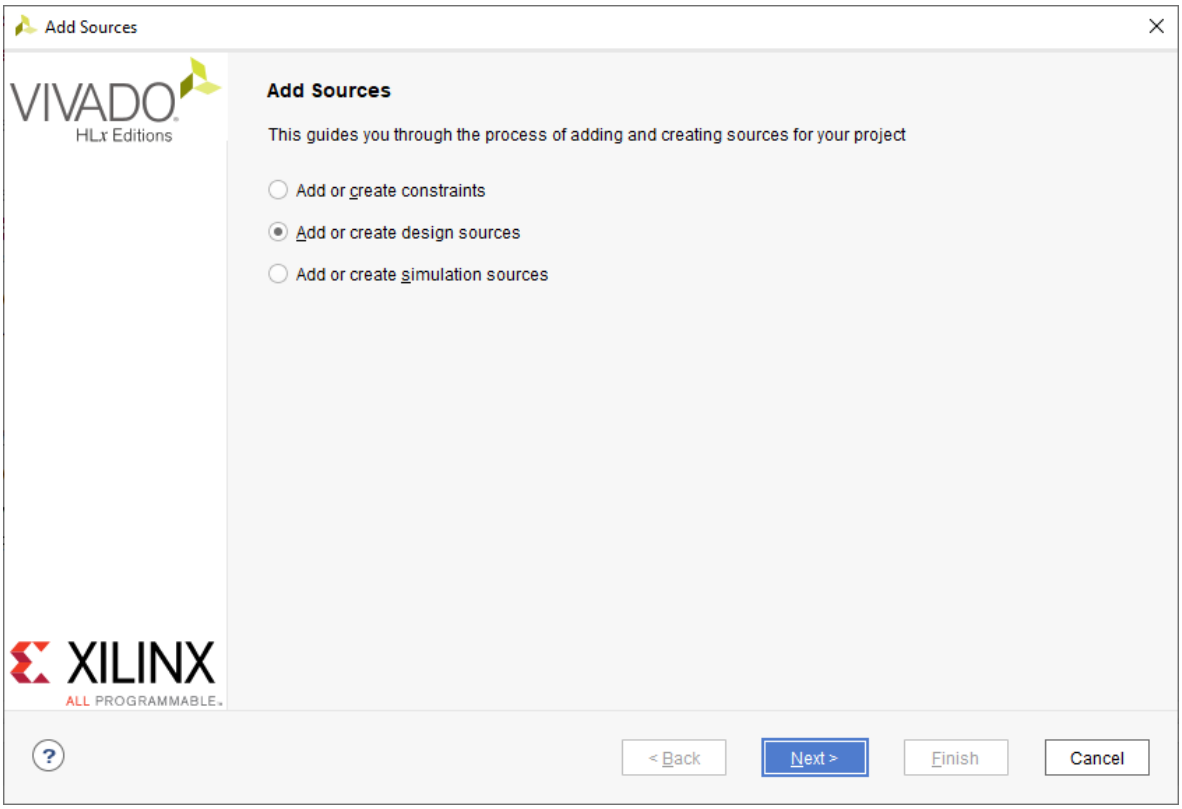
Add Ram Sources to Project

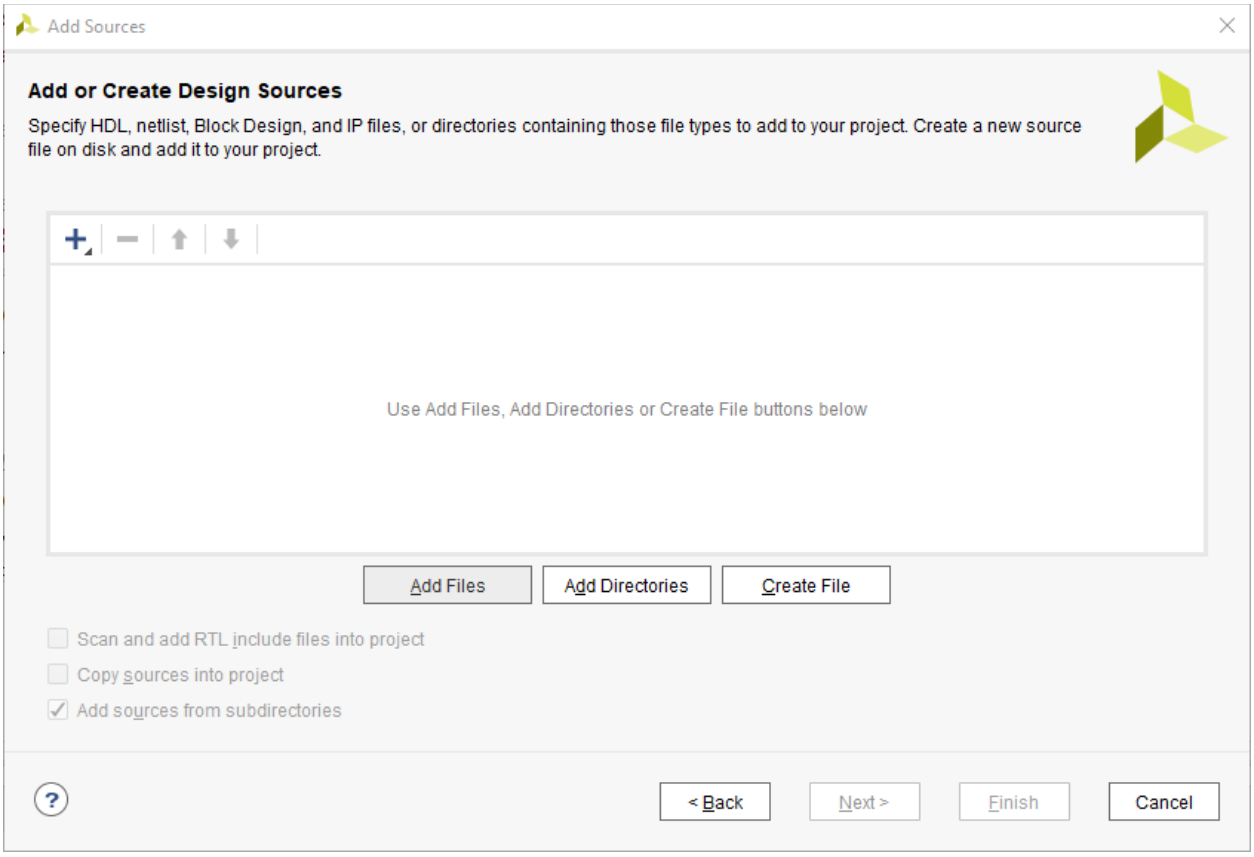
Click the Add Sources under Sources Tab.



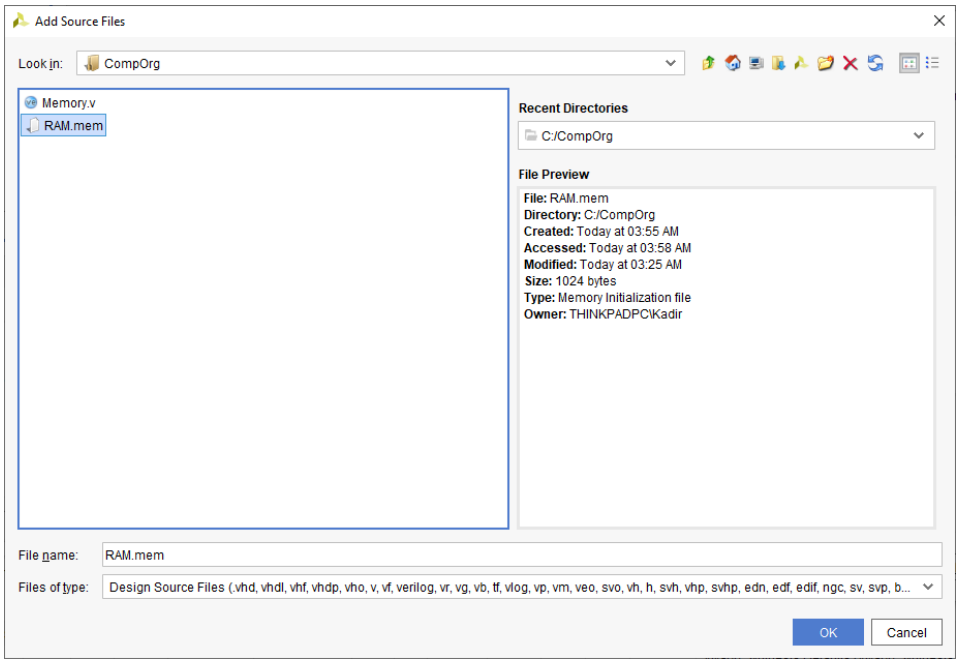
Select “Add or create design sources”, then click the next button.



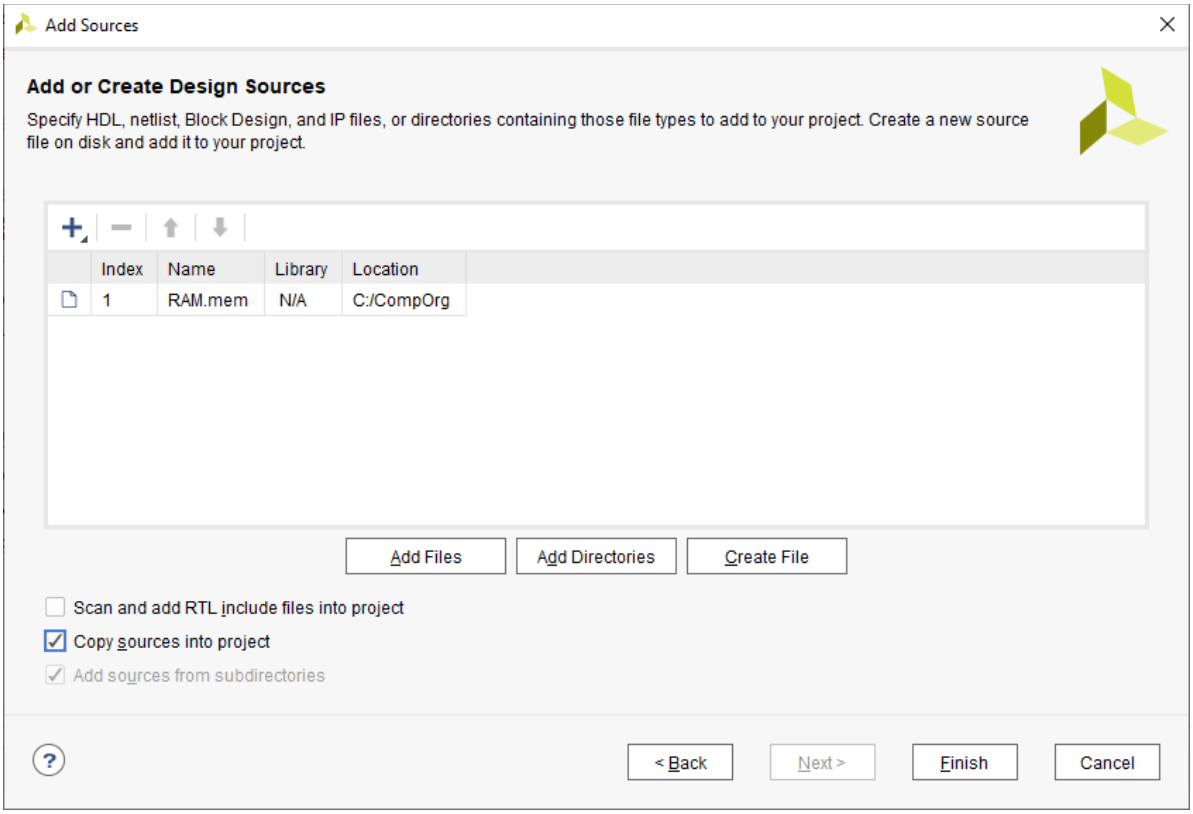
Click “Add Files” button.



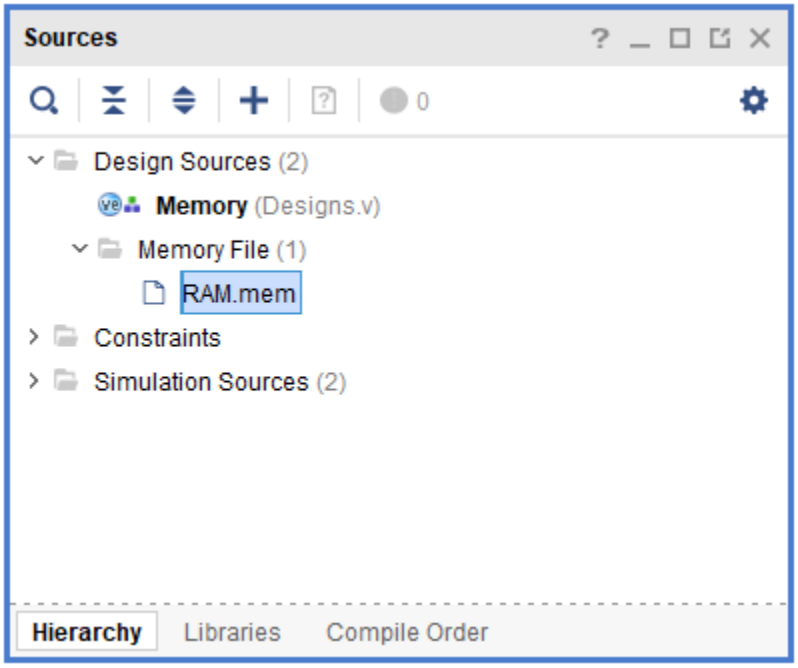
Select the RAM.mem file from File Explorer. Then click OK button.



Select the “Copy sources into project” option and click Finish button to add file.



RAM file can be accessed under sources section.

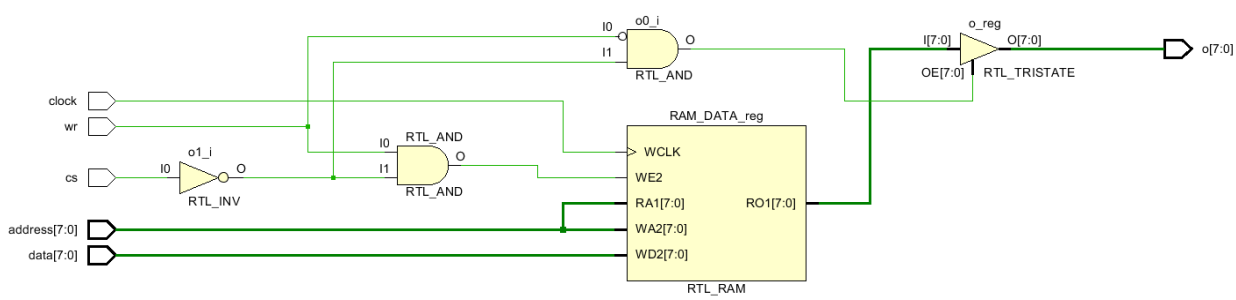


The RAM module is given in DataMemory.v file.

```
module DataMemory(
    input wire[15:0] Address,
    input wire[7:0] Data,
    input wire WR, //Read = 0, Write = 1
    input wire CS, //Chip is enable when cs = 0
    input wire Clock,
    output reg[7:0] MemOut // Output
);
    //Declaration of the RAM Area
    reg[7:0] RAM_DATA[0:65535];
    //Read Ram data from the file
    initial $readmemh("RAM.mem", RAM_DATA);
    //Read the selected data from RAM
    always @(*) begin
        MemOut = ~WR && ~CS ? RAM_DATA[Address] : 8'hZ;
    end

    //Write the data to RAM
    always @(posedge Clock) begin
        if (WR && ~CS) begin
            RAM_DATA[Address] = Data;
        end
    end
endmodule
```

You can access your RAM using this module.



Add ROM.mem as a source file to the project in the same way as RAM.mem.

The ROM module should be written in the InstructionMemory.v file.

Content of the RAM.mem and ROM.mem files: Each line shows the initial content of the memory cell. For Project 1, all values are random. These values in ROM will represent instructions for Project 3.

```
00
00
00
00
00
00
00
00
00
00
00
00
00
00
00
00
```